HAMZA MEZZENE

Network & Cybersecurity Engineering Student

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PROFESSIONAL SUMMARY

Third-year cybersecurity engineering student at UIR (Ing3, expected 2028), with hands-on experience in network configuration, Linux system security, and CTF competitions. Deployed multi-segment networks using iptables NAT, BIND9 DNS, and dnsmasq DHCP inside isolated Linux namespaces. Active CTF player specializing in steganography and cryptanalysis — co-organized CYBrAIn Fest UIR, a two-day AI × cybersecurity event for 40+ participants. Certified by Cisco in Cyber Threat Management and Introduction to Cybersecurity. Looking for a cybersecurity or network engineering internship where real infrastructure and security depth matter.

TECHNICAL SKILLS

Cybersecurity: Network Security, Penetration Testing (fundamentals), CVSS scoring, Incident Response, Cyber Kill Chain, Diamond Model, Threat Intelligence, CTF (Steganography, Cryptography)

Networking: DNS/BIND9, DHCP (dnsmasq), NAT/iptables (MASQUERADE, SNAT), Linux Network Namespaces, TCP/IP stack, Wireshark (packet analysis), VirtualBox (lab environments)

Operating Systems: Linux (Ubuntu) system administration, namespace isolation, firewall rules, service configuration

Data / BI: PySpark, BigQuery, Talend ETL (tMap, tAggregateRow)

Languages: Python, Bash, PHP 8, Java (Android), SQL

Web & Backend: PHP MVC (custom), PDO, MySQL, AJAX/Fetch API, bcrypt, Session Auth, REST principles

Tools: Git/GitHub, VSCode, Wireshark, VirtualBox

CTF & SECURITY PRACTICE

CTF Challenges — Steganography, Forensics & Cryptanalysis

Apr 2026

CYBERGLITCHERS Club

Python · LSB Steganography · Cryptanalysis · Binary Forensics · Frequency Analysis

- Extracted hidden payloads from PNG files using LSB (Least Significant Bit) analysis — located embedded data across image channels.
- Reverse-engineered obfuscation transformations with a custom brute-force decryption pipeline written in Python.
- Broke classical ciphers (Caesar, Vigenère, columnar transposition) using frequency analysis and known-plaintext attacks.
- Applied binary forensics techniques to identify file signatures, extract metadata, and locate concealed artifacts.

PROJECTS

Linux Networking Lab — Multi-Segment Network Architecture

Apr 2026

Networking Module, UIR

Linux · BIND9 · iptables · dnsmasq · Network Namespaces · Bash

- Designed and deployed an isolated multi-segment network topology using Linux network namespaces — each namespace operating as an independent routing domain.
- Configured authoritative DNS with BIND9: zone files, forward/reverse lookup, and resolution across namespaces.
- Implemented NAT routing with iptables — MASQUERADE for dynamic outbound traffic, SNAT for static address translation between segments.
- Deployed DHCP with dnsmasq for automated IP assignment, lease management, and DNS integration in lab environments.
- Validated end-to-end connectivity with packet-level tracing and firewall rule inspection.

Stocky — Multi-Role Commercial Platform

Mar 2026

Web Development Module, UIR

PHP 8 · MySQL/PDO · AJAX/Fetch API · bcrypt · Custom MVC · Session Auth

- Built a full-stack web application from scratch with no framework — custom MVC, single front controller, role-scoped routing (Admin, Vendor, Supplier, Client).
- Implemented secure authentication: bcrypt password hashing, session management, and role-based access control.
- Admin dashboard with real-time revenue tracking, stock alert thresholds, and user management.

EXPERIENCE & LEADERSHIP

Vice-President *2024 – Present*

CYBERGLITCHERS UIR — Cybersecurity Club

- Lead club coordination: CTF event organization, technical workshops, and external partnerships.
- Managed all logistics and cross-team communication for CYBrAIn Fest (2-day event, 40+ participants, 3 partner organizations).
- Oversaw speaker management, scheduling, and on-site operations.

Sponsor Manager *Jan 2026 – Present*

The Crew BDE UIR — Student Union

- Handle sponsor outreach and partnership negotiations for student campus events.
- Coordinate between union leadership and external companies for funding and visibility agreements.

Operations Volunteer — AFCON 2025 *Dec 2025 – Jan 2026*

OLM Souissi Fan Zone, Rabat

- Coordinated team assignments and real-time logistics for a large-scale international event.
- Made on-the-spot operational decisions under time pressure with multiple concurrent teams.

EDUCATION

Engineering Degree — Cybersecurity (Ing3, expected 2028) *2025 – 2028*

École Supérieure d'Informatique et du Numérique (ESIN) — Université Internationale de Rabat (UIR)

- Currently in third year (Ing3). Core modules: Network Security, Cryptography, Linux Networking, Web Development, Mobile Development (Android), Databases & BI, Software Engineering.
- Vice-president of CYBERGLITCHERS, the university's cybersecurity club. Active CTF organizer.

CERTIFICATIONS

Cisco — Cyber Threat Management · CVSS, Incident Response, Penetration Testing, Threat Intelligence · 2024

Cisco — Introduction to Cybersecurity · Network Defense, Threat Analysis, Asset Management · 2024

Coursera — Data Analysis with PySpark · 2023

Coursera — Working with BigQuery · 2023

LANGUAGES

Arabic — Native **French** — B2+ **English** — B2